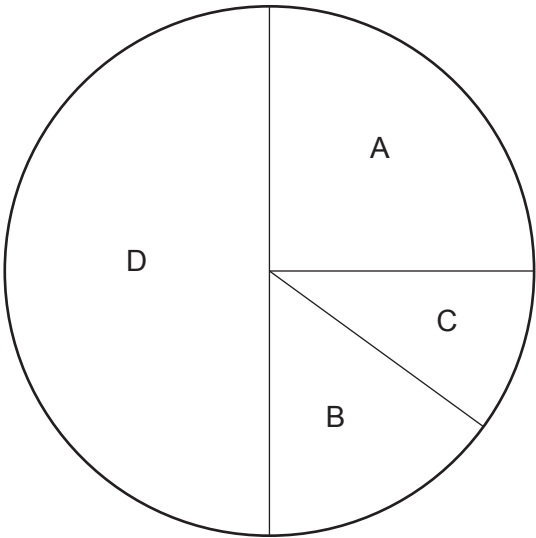


5. (a) The pie chart shows the percentages of background radiation from various sources for an area of Cornwall in the UK.

(i) Use the following information to **complete the key** for the pie chart. [3]

Around 50 % of background radiation comes from radon gas.
25 % comes from ground and buildings and medical together.
15 % comes from cosmic rays.
10 % comes from food and drink.



Key for pie chart

Source of background radiation	Letter on pie chart
Radon gas	
Food and drink	
Ground and buildings and medical	
Cosmic rays	

(ii) The background radiation from radon gas in Mid Wales is **smaller** than in Cornwall.
The background radiation from all other sources remains the same.

Alan states that the pie chart for Mid Wales would be the same as the one above.
Explain whether you think that Alan is correct. [2]

.....

.....

.....



(b) The isotope polonium-210 has a half-life of 140 days.
It is a source of alpha radiation.
200 g of polonium-210 is stored in a locked cupboard.

(i) The mass of the polonium-210 halves in 140 days.
State **one** other quantity that halves in that time.

[1]

(ii) **Complete the table** below to show how the mass of polonium remaining changes with time.

[2]

Time (days)	Mass of polonium remaining (g)
0	200
140	100
280	
420	

(iii) **Complete the following table**, giving the numbers of particles in a nucleus of polonium-210 ($^{210}_{84}\text{Po}$).

[3]

Name of particles	Number of particles in the nucleus
nucleons	
protons	
neutrons	

11

